Real-Time Crowd Density Estimation Using Self-Distilled ResNet-50 Networks

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***Abstract*—Crowd counting is a basic problem of computer vision that has a wide range of applications including public safety, urban management, and video surveillance through density map estimation. Although the recent efforts based on deep learning have resulted in significant accuracy, a lot of them are based on complex multi-column architectures or external teacher-student frameworks, which ultimately cause the increased peripheral computation overhead and training complexity. In order to overcome these limitations, we propose a self-distilled crowd density estimation framework based on one ResNet-50 backbone, which avoids the use of external teacher by adopting the implicit teacher, exponential moving average (EMA) to serve as an implicit teacher, which allows the student network to effectively transfer its knowledge within the same architecture. The process is trained with a composite loss comprising pixel-wise density supervision, global count consistency, and the self-distillation regularization guided loss that ensures stable optimization and better generalization. The proposed framework is validated on the ShanghaiTech Part-A and Part-B benchmarks, on both dense and sparse crowd by using unified architecture, and the experimental results achieve competitive results while keeping an architectural simplicity. Specifically, the model gives Mean Absolute Error (MAE) of 93.87 for ShanghaiTech Part-A and 15.01 for ShanghaiTech Part-B, which points to its effectiveness, especially for sparse crowd settings. Overall, the proposed approach provides a good balance between the accuracy, model simplicity, and computational efficiency, which is suitable for practical crowd counting applications without any external teacher models.**

***Index Terms*—Crowd Counting, Density Map Estimation, Self-Distillation, EMA Teacher, ResNet-50, Deep Learning**

I. Introduction

Crowd counting is one of the primary challenges in computer vision that has important applications in the area of public safety monitoring, urban planning, traffic control, and global event management. For example, accurately estimating the crowd size in highly-congested scenes is still a challenging task because of the severe occlusions, perspective errors, scale variations, and crowd distribution. Early approaches were mostly based on the detection-based and regression-based approaches, however, these approaches also often break down in dense scenarios, where the individual detection becomes unreliable or infeasible [1, 2].

To circumvent these shortcomings, an alternative paradigm, density map estimation, has become the prevailing paradigm in modern crowd counting systems. Rather than detecting individual pedestrians, density-based approaches based on continuous spatial representation study a function in which the integral of the density map represents the total number of people of a crowd. The advent of convolutional neural networks (CNNs) greatly improved this paradigm and multi-column architectures like MCNN [1] were designed for coping with scale variation by processing features with different receptive fields. Despite their effectiveness, multiple-column networks generally have high computation costs and complicated training pipelines.

Subsequent research was dedicated to improving feature representation and scale robustness as well as attempts to reduce the architecture complexity. CSRNet [3] used dilated convolutions to increase receptive fields without adding parameters, and achieved good performance in highly crowded scenes. Other methods involved the scale aggregation [4], distribution matching [5] and the iterative refinement strategies [6]. Adaptive and switch-based architectures dynamically chosen feature extractors based on the crowd density [7, 9]. Although these approaches increased accuracy, they were often accompanied by increasing the number of modules and adding extra complexity to the design, so they were often not practical for real-time implementation.

More recently, studies have focused on efficiency and light weight design of models. Approaches such as LiteCrowd [10] and context-aware networks [11] attempt to strike between performance and low computational overhead. Semi-Supervised, Ranking-based, and Weakly-Supervised methods further try to enhance generalization using few annotations [12, 18]. Nevertheless, there are still many state-of-the-art solutions that rely on sophisticated architectures, multi-branch architectures, or auxiliary supervision, which makes them less suitable for deployment in resource constrained or real-time environments.

Knowledge distillation has been studied as an approach to enhance lightweight models by transferring knowledge from high-capacity teacher networks [17]. In crowd counting, distillation-based frameworks usually need to train and maintain an extra teacher model, which leads to more memory consumption, and training challenge. Recently, self-distillation techniques are suggested to share knowledge among the same network between the stages of training process [16]. While promising, the study of self-distillation for efficient and unified crowd density estimation is still relatively under-explored.

Motivated by these observations, a self-distilled crowd density estimation framework of a single ResNet-50 architecture has been proposed in this paper. Instead of using an external teacher, the proposed method uses an exponential moving average (EMA) of the student network as an implicit teacher, which allows the self-knowledge transfer to be stable and efficient. The model is optimized with a compound objective combining pixel-wise density supervision, global count consistency, and self-distillation regularization (EMA based). Notably, the same architecture is used for both dense and sparse crowd scenarios, ensuring the architecture simplicity and efficiency in deployment.

The proposed approach is evaluated on the ShanghaiTech Part-A and Part-B datasets which show competitive performance, with a significant reduction in the complexity of the architecture and training. By omitting multi-branch designs and outside teacher networks, the proposed framework gives a suitable and effective answer for crowd counting frameworks based on deployment

II. Related Work

Early methods for counting the crowd were based on the visual features that was handcrafted and on traditional machine learning methods. Using these methods, different descriptors were extracted such as Scale-Invariant Feature transform (SIFT), histograms of oriented gradients (HOG), local binary pattern and further the different kind of the regression or classification models were implemented to infer the total number of people present in a scene. While having been successful in sparse environments, such techniques resulted in huge performance deterioration in dense scenes in terms of severe occlusions, perspective distortions and scale variations [1, 2]. The representational ability of hand-crafted features was also limited and therefore their application in real-world dense crowed scenarios was limited.

The advent of convolutional neural networks (CNNs) contributed greatly to the improvement of crowd counting since it boosts the ability of automatic feature learning and hierarchical representation extraction. A seminal work in this direction is the Multi-Column Convolutional Neural Network (MCNN), which introduced the use of parallel convolutional branches in order to explicitly account for scale variation [1]. By re-formulating the crowd counting problem as a task of density map regression, MCNN was able to give spatial distribution information besides accurate count information. Building on this idea, CSRNet used a VGG-16 backbone with dilated convolutions for the purpose of increasing the receptive fields without changing the spatial resolution and achieved good performance in extremely congested scenes [3]. Dilated convolution-based architectures have since become a very popular kind of architecture to use as a benchmark for crowd density estimation.

Subsequent studies were focused on improving the robustness of scales, and on modeling taking place from a contextual point of view. Scale aggregation networks were mixing multi-scale features to better performance for different crowd densities [4]. Distribution matching approaches proposed the introduction of the loss functions, which ensures the matching of the predicted and the ground truth density distributions, improving further the accuracy of the estimation [5]. Attention mechanisms and context-aware designs also were introduced to suppress the background noise and emphasize the crowd relevant regions [7, 8]. Although these methods made some significant improvements, they often included several other architectural elements and complicated training strategies that added a greater computational cost and rendered them inappropriate for real-time deployment.

In parallel with this there was an attempt to try to make the computes efficient, and to minimize the complexity of the model. Lightweight crowd counting models were proposed for resource constrained platforms. Approaches such as LiteCrowd looked at efficient convolutional designs to reduce the number of parameters without any major alteration in the accuracy [10]. Single-column variants of multi-column networks tried to remove the redundancy in the network and in the inference would be much faster with small performance tradeoffs [8, 9]. Due to their stable optimization and high representational capability, residual networks also have become popular. ResNet-based backbones with light-weight regression heads are a good trade-off between accuracy and efficiency with density estimation tasks.

Knowledge distillation has been a popular approach to improve compact models by transferring the knowledge from a large capacity teacher network [17]. Traditional production frameworks for distillation are based on a teacher-student paradigm (student learns to imitate the teacher's predictions or intermediate representations). While they work for a variety of vision tasks, these approaches add extra training complexity and memory overhead for having to have another teacher model. To bypass these pitfalls, the concept of self distillation approaches has been introduced which enable a model to learn from its own predictions or history [16]. Self-distillation is a mechanism for regularization of prediction consistency and better generalization without external supervision.

Recent efforts were devoted to self-distillation in crowd counting, particularly in some special cases such as domain adaptation and continual learning [16]. However, some of the currently existing methods are based on complex training strategies or auxiliary supervision signals. Moreover, combining self-distillation and real-time estimation of crowd density in unified and deployment-oriented architecture still requires to be too much explored. In contrast, the proposed approach takes self distilled crowd density estimation framework which is based on single ResNet 50 architecture. By using an exponential moving average (EMA) of the student network as an implicit teacher the model enjoys stable self-supervision whilst keeping the architecture simple. This design eliminates the need for external teacher models and multi-branch architectures, resulting in an accuracy and computational efficiency trade off in previous work.

III. Methodology

Within this section there is the complete outline of self distilled real time crowd density estimation. The proposed There are four major components of approach as follows: (1) dataset prepa ration and preprocessing, (2) unified ResNet-50 network archi tecture for density estimation, (3) formulation

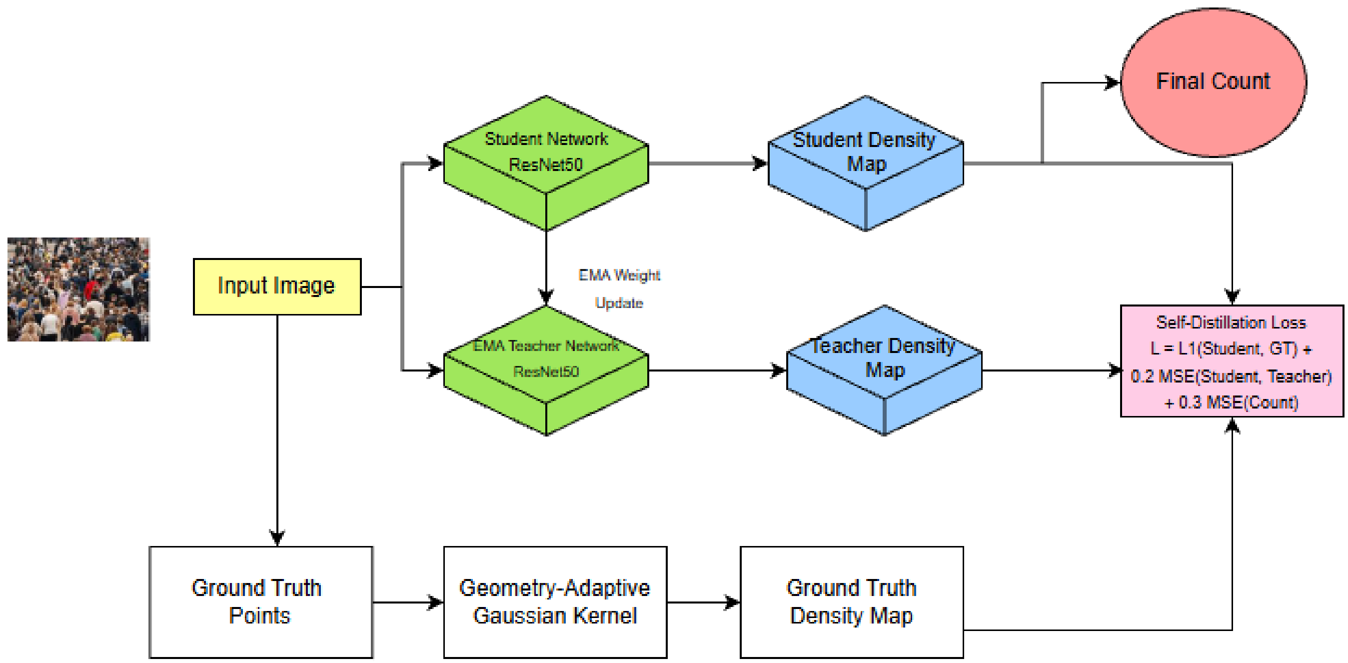

Fig. 1: Overview of the proposed self-distilled ResNet-50 architecture for crowd density estimation.

Algorithm 1 Self-Distillation Training for Crowd Density Estimation

Require: Training images X, ground-truth density maps D_{gt}
Ensure: Trained self-distilled ResNet-50 model

```
Initialize student network S with ImageNet-pretrained weights
Initialize EMA teacher T ← S
for each epoch do
    for each mini-batch (x, D_gt) do
        D_stu ← S(x)
        D_ema ← T(x)
        Compute L_sup using D_stu and D_gt
        Compute L_count using integrated density values
        Compute L_sd between D_stu and D_ema
        L_total = αL_sup + βL_count + γL_sd
        Update student parameters using backpropagation
        Update EMA teacher: T ← μT + (1 − μ)S
    end for
end for
return Trained student model S
```

of composite self-distillation loss function., and (4) training and evaluation., and (4) training and evaluation., and (4) training and evaluation., and (4) training and evaluation., and (4) training and evaluation., and procedures. As opposed to traditional teacher-student structures, the method proposed is self-distillation using an expo nential moving average (EMA) teacher based on the same network, therefore eliminating the need for an outside teacher model without losing stability efficiency during training

A. Dataset and Preprocessing

The proposed approach is evaluated on ShanghaiTech dataset with two subsets containing significantly different characteristics of the crowded. ShanghaiTech Part A consists of 482 images downloaded from the Internet with extremely dense crowds and B consists of 716 images acquired in urban street scenes with sparse to moderate densities. After following the traditional evaluation rules, 300 images are used for the training purpose from Part A and 400 images are used from part B and few images are kept to test the accuracy. Each image is annotated with the head center coordinates, which makes it possible to perform precise supervision of the density maps.

Density maps for ground truth are generated by placing geometry-adaptive gaussian kernels at the point of each annotated head. The standard deviation of each Gaussian is represented depending on the average distance to its nearest neighboring annotations, except for itself, thus effective adaptive kernel size to local crowd density. In such an approach, areas with high density create wider kernels whereas areas with low density produce thinner kernels. All density maps are normalized in order to have the integral over the spatial domain equal to actual crowd count. In the training process, images are randomly cropped into fixed-size patches that can be processed in batches, and the intensity of the pixels is normalized, with the statistics of ImageNet as the statistics.

B. Network Architecture

The network architecture is built on the one ResNet-50 backbone pretrained on ImageNet. Its backbone is based on an initial convolution and a pooling layer and four subsequent stages of residual processing which sequentially encode hierarchical representations of features, but stabilize the training process using identity skip connections. The generated final feature maps have the resolution of a spatial scale of about 1/32 of the input size. A decoder built of a series of convolutional layers is connected at the backbone end to decrease dimensionality in channels and regress a single-channel density map. To recover the spatial resolution, bilinear upsampling is used to make sure that the sum of the predicted density map is equal to the predicted crowd size. The single-column design has the advantage of not causing architectural redundancy and is real-time deployable.

To train the model, a composite self-distillation loss is employed. Let D_{gt} denote the ground-truth density map, D_{stu} the student prediction, and D_{ema} the EMA teacher prediction. The supervised density loss is defined as a pixel-wise ℓ_1 loss:

$$L_{\text{sup}} = \frac{1}{N}\sum_{i=1}^{N} \|D_{stu}^{(i)} - D_{gt}^{(i)}\|_1. \tag{1}$$

To enforce global count consistency, a count loss penalizes discrepancies between predicted and ground-truth counts:

$$L_{\text{count}} = \frac{1}{N}\sum_{i=1}^{N}\left(\sum_{x,y} D_{stu}^{(i)}(x,y) - \sum_{x,y} D_{gt}^{(i)}(x,y)\right)^2. \tag{2}$$

Self-distillation regularization is achieved by minimizing the mean squared error between student and EMA teacher predictions:

$$L_{\text{sd}} = \frac{1}{N}\sum_{i=1}^{N} \|D_{stu}^{(i)} - D_{ema}^{(i)}\|_2^2. \tag{3}$$

The total training objective is defined as:

$$L_{\text{total}} = \alpha L_{\text{sup}} + \beta L_{\text{count}} + \gamma L_{\text{sd}}, \tag{4}$$

where α, β, and γ balance local density accuracy, global count fidelity, and self-distillation regularization. The EMA teacher parameters are updated using:

$$\theta_{ema} \leftarrow \mu\theta_{ema} + (1-\mu)\theta_{stu}, \tag{5}$$

where μ denotes the momentum coefficient.

The training is conducted in a single-stage process with the use of. AdamW optimizer. The propagation of gradients is only done on the basis of. the student network, the EMA teacher is informed through. momentum-based averaging. Only the stu- does it have during inference. This is done using dent network, which does not involve extra computation. overhead. Shang- Performance Model Model performance is measured on the Shang- The test sets in haiTech are based on Mean Absolute Error (MAE) and Root. Mean Squared Error (RMSE)..

C. Training Loss Curve and MAE Trends

Figure 2 and Figure 3 demonstrates the dynamics of Mean Absolute Error (MAE) during training epochs of the proposed self-distilled ResNet-50 model on ShanghaiTech Parts A and B, respectively, to learn how EMA-based self-distillation converges and becomes stable. In the case of ShanghaiTech Part

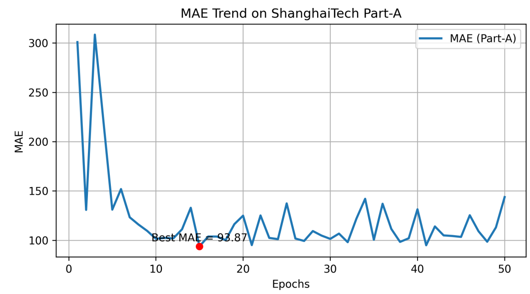

Fig. 2: MAE trend over epochs on ShanghaiTech Part A.

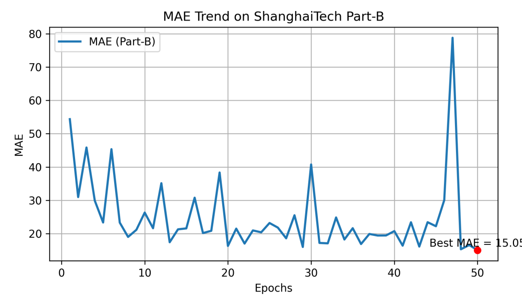

Fig. 3: MAE trend over epochs on ShanghaiTech Part B.

B, MAE goes down steeply in the first epochs, and this means that coarse density patterns are quickly learned in sparse and moderately dense scenes. The error subsequently stabilizes and approaches lower values, and this proves that the EMA teacher has valuable regularization, which inhibits the occurrence of noisy predictions and enhances generalization. ShanghaiTech Part A on the contrary has increased MAE variations because of excessive crowd density, extreme occlusions, and variations in size. Spatial inaccuracies in dense parts are minor and may cause substantial effects on total count estimation, and the slowly changing EMA teacher adds short-term variations in validation error. However, the general direction is of convergence with the optimum checkpoint with significantly lower error than the early epochs. The above findings indicate the significance of validation-based checkpoint selection and that EMA-based self-distillation can be effectively and stably trained in densely and sparsely crowd scenarios.

D. Model Efficiency and Trade-offs

The self-distilled ResNet-50 suggested framework is self-distilled and efficient and can be deployed in practice. The proposed method uses an exponential moving average (EMA) of the student itself, rather than another network of high capacity teacher, unlike traditional teacher-student distillation methods that are forced to use an extra teacher during inference.

Computationally speaking, inference is a priori of the same cost as a regular ResNet-50-based density estimator because no additional parameters or auxiliary branches are required during test time. This allows the model to be used in real-time and it is much simpler to architecture than multi-column or attention-intensive networks.

TABLE I: Comparison of crowd counting methods on ShanghaiTech dataset (lower MAE indicates better performance).

Method	Backbone	MAE (A) ↓	MAE (B) ↓
MCNN [1]	Multi-column CNN	110.2	26.4
CSRNet [3]	VGG-16 + Dilated	68.2	10.6
SANet [4]	Scale Aggregation	67.0	8.4
Distribution Matching [5]	VGG-16 + DM	65.6	7.8
SwitchCNN [7]	Multi-Switch CNN	90.4	21.6
LiteCrowd [10]	Efficient CNN	79.0	12.4
SPACNet [8]	Pyramid Attention	62.4	7.0
Proposed Method	**Self-Distilled ResNet-50**	**93.87**	**15.05**

The major trade-off is manifested in very high density crowd scenes. like ShanghaiTech Part A, in which multi-scale specialization is applied. architectures can also be lower MAED with higher. calculation and decreased flexibility in implementation. In contrast, the suggested approach is less complex, more efficient, and generalization, which presents a middle way solution to real world. crowd counting applications.

IV. Results and Discussion

A. Quantitative Performance

Table II provides a quantitative comparison of the suggested self-distilled ResNet-50 model and various representative methods of counting crowds on the ShanghaiTech dataset. The comparison is done based on Mean Absolute error (MAE) in which low values represent the improvement of counting.

The findings indicate that the suggested approach has MAE of 93.87 on ShanghaiTech Part A and 15.05 on Part B. Some of the state-of-the-art solutions have lower MAE on Part A, but generally, these are based on multicolumn architectures, explicit scale-sensitive modules, or external teacher-student designs. Conversely, the suggested methodology utilizes one ResNet-50 architecture with self-distillation via EMA, which makes the model much simpler to design and train.

It is especially remarkable in terms of the performance on ShanghaiTech Part B. The suggested self-distillation strategy is advantageous to sparse and moderately dense scenes, with the result of low MAE 15.05. This shows that self-supervisory EMA is largely effective in regularizing a network and enhancing generalization in situations where there is low crowd density and greater spatial distance between individuals.

On the whole, the quantitative findings indicate that the proposed framework offers a competitive trade-off between accuracy and architectural simplicity, and thus it is suitable in real-time and resource-constrained tasks in the crowd counting.

B. Qualitative Results

Figures 4 and 5 compare qualitative results of ground-truth versus predicted density maps on ShanghaiTech Part A and Part B respectively. Both figures depict the image of the input and the respective ground-truth density map and the predicted density map by the proposed model.

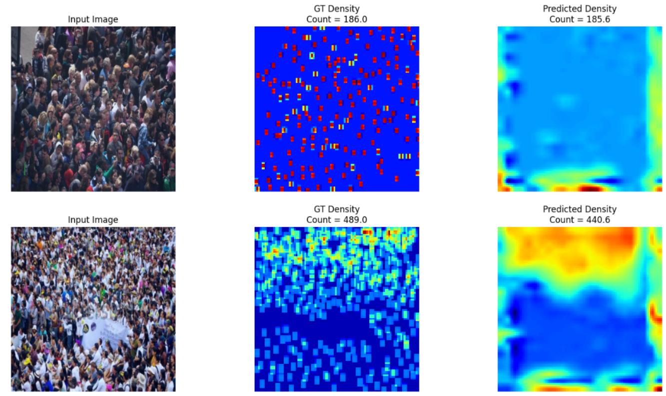

Fig. 4: Qualitative results on ShanghaiTech Part A. From left to right: input image, ground-truth density map, and predicted density map.

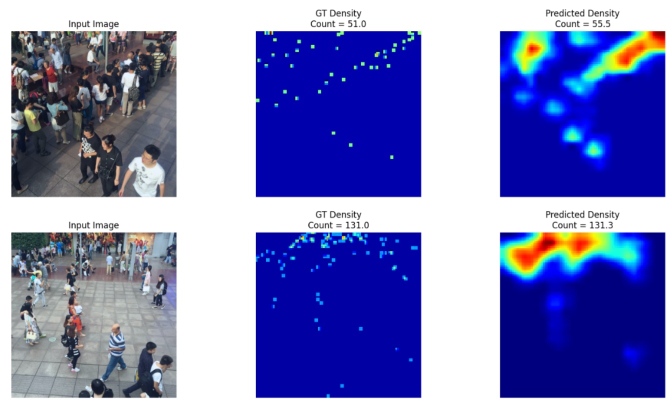

Fig. 5: Qualitative results on ShanghaiTech Part B. From left to right: input image, ground-truth density map, and predicted density map.

In ShanghaiTech Part A, where the density of the scenes is extremely high, the density map that is predicted successfully depicts the high-density areas and the general distribution of the crowd. Despite the fact that fine-grained separation between heavily occluded individuals is difficult, the predicted maps have global density structure, and generate reasonable total count estimates.

In the case of ShanghaiTech Part B, strong localization ability is attested by the qualitative results. The model finds individual clusters correctly and does not overactivate the background, which leads to clean and spatially coherent density maps. This qualitative behavior goes in line with the low MAE seen quantitatively in Part B and points to the usefulness of self-distillation in the sparse crowd setting.

By and large, the qualitative findings validate the hypothesis that the self-distilled ResNet-50 model yields smooth, interpretable, and local density maps with respect to diverse crowd densities to support the quantitative results provided above.

V. Conclusion and Future Scope

In this paper, a self-distilled crowd density estimation framework was presented on the basis of a single ResNet-50 framework that is intended to be deployed in real-time. In contrast to the traditional methods based on the usage of complex multi-column networks or external teacher-student distillation pipelines, the proposed system uses an exponential moving average (EMA) of the student network as an implicit teacher to facilitate effective knowledge transfer within the student network with less training complexity and memory. This model is trained on a compound loss based on pixel-wise density supervision paired with global count consistency and self-distillation regularization, which leads to stable training and better generalization. The experimental findings on the ShanghaiTech benchmark prove that using the same ResNet-50 architecture, a high-quality performance can be achieved in both dense and sparse crowds, with MAE of 93.87 on Part A and 15.05 on Part B, and the model keeps its architectural simplicity and substantial inference speed.

Future research will involve the additional improvement of efficiency and strength by integrating lightweight attention systems and adaptive self-distillation. It will also explore extensions to spatio-temporal modeling of video-based counting of crowds in order to take advantage of temporal consistency across frames. Moreover, pruning and quantization methods of model compression can be explored to lower the inference latency and allow implementation on minimal edge devices with resource constraints.

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